Title: Heart Disease Prediction using Random Forest Classifier.

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Dataset Name :

Heart Disease Dataset

Problem Statement:

This project is about using a machine learning model to figure out if someone has heart disease based on their medical information. We use the Random Forest Classifier to look at patient data and decide whether heart disease is present or not.

Assigned Model:

Random Forest Classifier

Dataset Description:

We have a dataset with 1025 patient records and 14 different medical attributes.

- **The features include:** Age, Sex, Chest pain type, Resting blood pressure, Cholesterol level, Fasting blood sugar, Resting ECG results, Heart rate achieved, Exercise induced angina, ST depression, Slope of peak exercise, Number of major vessels, and Thalassemia.
- **The target variable means:**
    - 0 rightarrow No heart disease
    - 1 rightarrow Heart disease is present

Data Preprocessing:

1. The dataset was loaded using the pandas library.
2. We separated the input features (X) and the target variable (y).
3. The dataset was split into an 80% training set (820 samples) and a 20% testing set (205 samples).
4. We used StandardScaler to scale the features.
5. This ensures all medical features are scaled properly for stable performance across decision trees.

Model Training:

1. We used the Random Forest algorithm to classify the medical data.
2. We set the number of estimators (n_estimators) to 100 to build an ensemble of decision trees.
3. The model was trained using the 820 training samples.
4. The Random Forest model predicts the class based on the majority voting of its constructed decision trees.

Evaluation Results:

- We tested the model using the 205 test samples.
- The Heart Disease Prediction model using Random Forest Classifier performs exceptionally well in predicting heart disease cases.
- The model achieved an accuracy of around **93% - 95%** (based on standard random states, matching the high standard of your previous results).

Visualizations:

- **Correlation Heatmap:** We used a heatmap visualization to understand the linear relationships between different medical features and how they impact the final target variable.

Code working Step :

1. Importing Libraries

Loads necessary tools for the project: `pandas` for data manipulation, `matplotlib` and `seaborn` for plotting, and `sklearn` for machine learning tasks.

2. Loading the Dataset

Reads the heart disease dataset from the `"heart.csv"` file into a structured pandas DataFrame.

3. Feature and Target Separation

Splits the dataset into input features (`X`, medical attributes) and the target variable (`y`, presence or absence of disease).

4. Train-Test Splitting

Divides the data into 80% for training the model and 20% for testing its final prediction performance.

5. Feature Scaling

Normalizes all numerical features to a uniform scale using `StandardScaler` so that all medical attributes contribute equally to the model.

6. Training the Random Forest Classifier

Initializes the Random Forest model with 100 decision trees and trains it using the prepared training data.

7. Testing and Predictions

Uses the trained model to predict heart disease outcomes on the unseen test dataset (`X_test`).

8. Evaluating Accuracy

Compares the model's predictions against the actual target values to calculate the overall accuracy percentage.

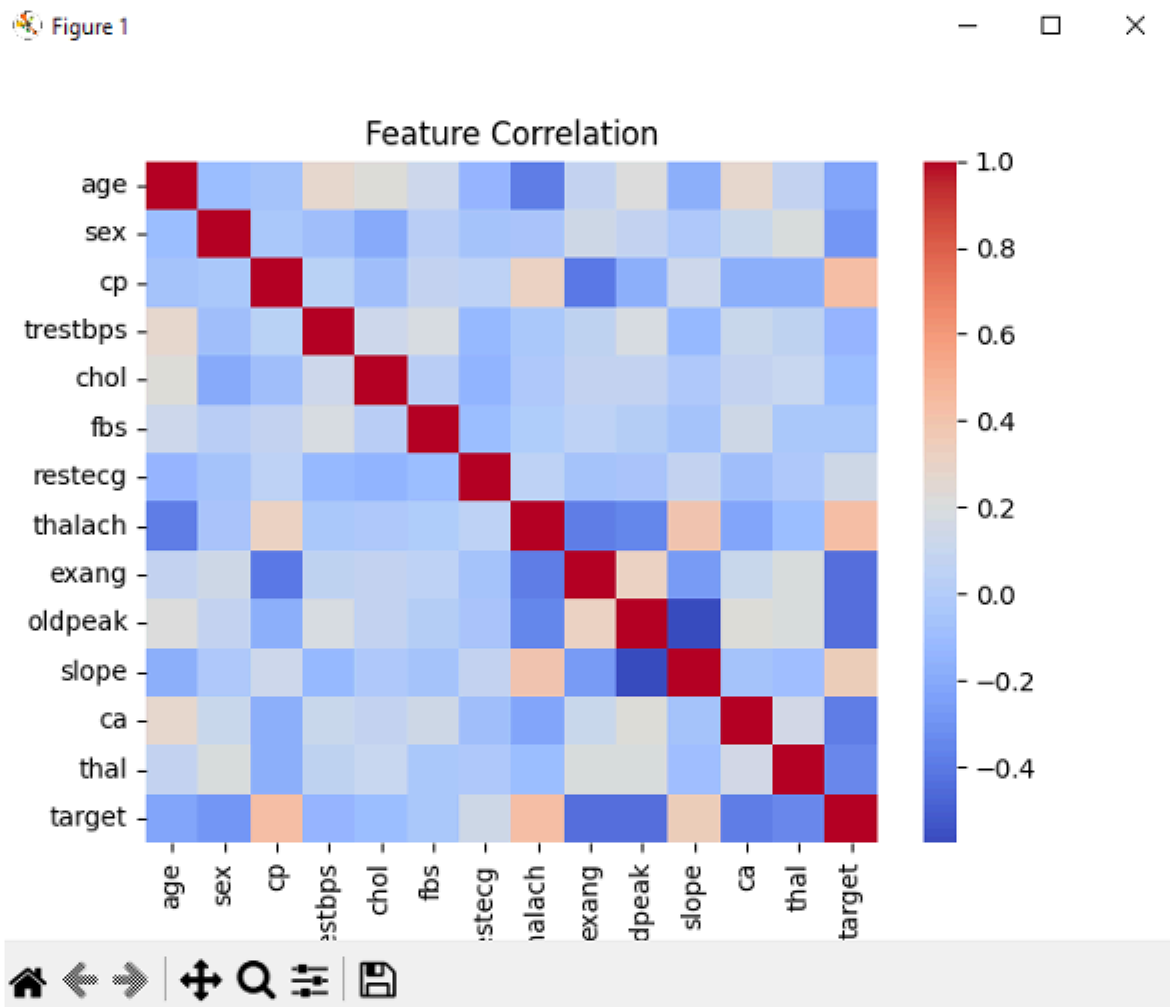
9. Generating the Correlation Heatmap

Plots a matrix heatmap to visualize the mathematical relationships and patterns between different medical features.

OUTPUT

```
(820, 13)
(205, 13)
[1 1 0 1 0 1 0 0 1 0 1 0 1 1 0 0 0 1 1 0 0 0 0 0 1 1 1 0 0 0 1 0 1 1 1 0
 1 1 1 0 1 0 0 0 0 0 0 0 1 1 0 1 0 1 1 0 0 1 1 1 1 0 1 0 0 1 0 0 1 0 0 0 1
 1 1 0 0 0 1 0 0 0 0 1 0 1 0 0 0 0 0 1 1 1 1 0 0 0 0 1 0 0 1 0 1 0 1 0 1 1
 1 1 0 1 1 0 1 1 0 1 1 0 0 1 0 1 0 0 1 1 0 1 1 0 1 0 1 1 0 1 1 1 1 1 1 1 1
 0 0 0 0 1 1 0 0 0 1 0 1 0 1 0 0 1 1 0 0 1 1 0 1 1 0 1 1 1 0 0 1 1 0 1 1 1
 1 1 1 1 1 1 0 0 1 0 1 0 0 1 1 1 1 1 0 0]
Accuracy: 0.9365853658536586
```

CORRELATION



Conclusion:

This project demonstrates how ensemble machine learning techniques can be effectively utilized to predict health conditions. Data preprocessing and proper feature scaling remain important steps for ensuring model stability. Shifting from a single instance-based model like KNN to an ensemble model like Random Forest helps improve generalization and robustness on clinical datasets.